AI FOR DATA ANALYSIS · LESSON 3 OF 4

AI in Excel & Office: Copilot, Gemini, Claude, Python in Excel & Agents

Use AI where analysts work every day. Work with Copilot in Excel (formulas, insights, charts, the COPILOT function), Gemini in Google Sheets and Excel add-ins, Claude and other assistants for formulas and files, Python in Excel, and office agents that perform multi-step tasks, and learn how to prompt, verify and stay safe.

● DVA with AI · class 3 ✦ Microsoft 365 Copilot, Gemini and agent features as of 2026

Intermediate ⏱ 120 min

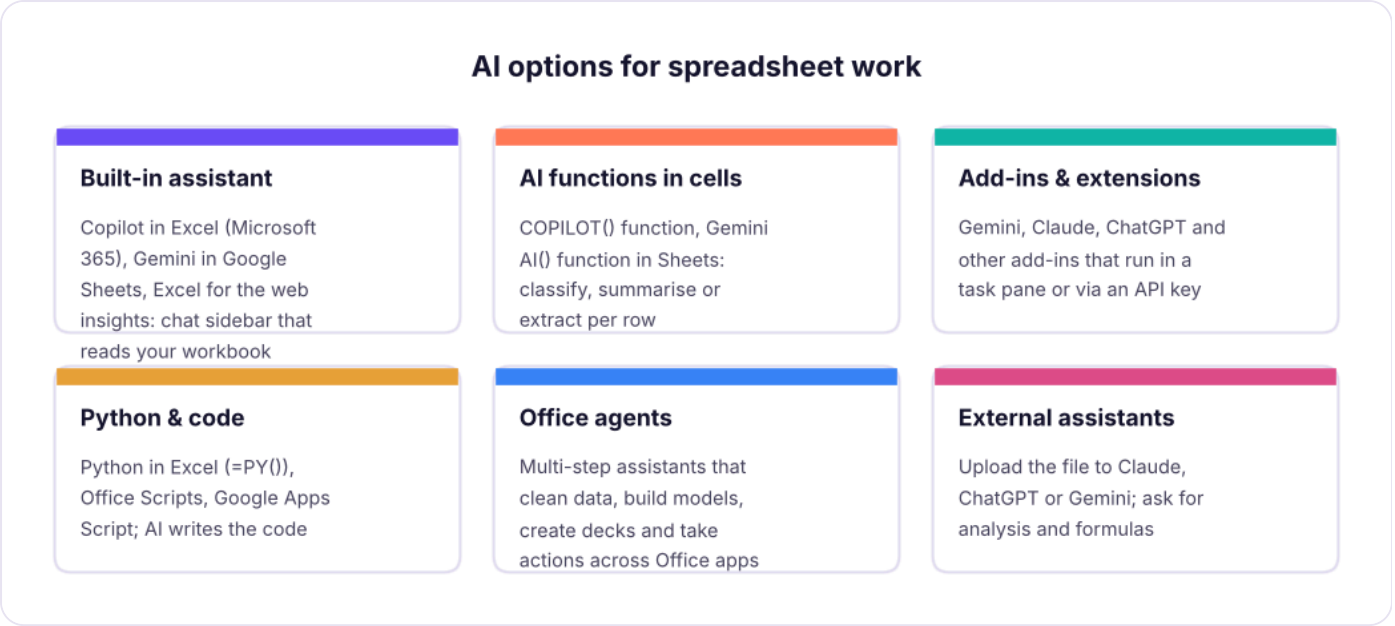
By the end of this lesson you can

- Prepare a workbook so AI tools understand it (Tables, clear headers, no merged cells)
- Use Copilot and Gemini-style assistants to explain, write and debug formulas, create pivots and charts, and summarise data
- Use the Excel COPILOT function and Python in Excel for text, classification and analysis inside cells
- Describe what office agents do and what controls they need
- Verify AI-generated formulas and numbers with independent checks

Datasets used

`superstore_sales.csv` `employees_small.csv`

The AI-in-Excel landscape



Four ways AI shows up in spreadsheets.

Step 0: make your workbook AI-ready

AI assistants can only reason about data they can *see and parse*. The same rules that make pivot tables work (Excel Lesson 1) make AI work:

1. **Convert data to an Excel Table** (`Ctrl + T`) with a clear name (`tblSales`) and one header row.
2. **Descriptive column names** (`order_date` , `sales_inr`) instead of `Col1` ; add units.
3. No merged cells, blank rows or notes inside the data; keep totals outside.
4. One sheet per dataset; a small **Data dictionary** sheet that defines columns and metrics (AI reads it too).
5. Remove or mask sensitive columns before connecting an AI service.
6. Save to **OneDrive/SharePoint** (Copilot needs cloud-saved files with AutoSave on).

Copilot in Excel

Needs a Microsoft 365 Copilot licence and your data formatted as a Table or range with headers.

Task	Example prompt
Explain	"Explain what column H does and how it is calculated"
Formulas	"Add a column that shows profit margin %" / "Write a formula that looks up the manager's name from the Managers table"
Insights	"Show me insights about sales by region" → Copilot adds pivot tables and charts
Highlight	"Highlight rows where discount > 20% and profit is negative" (conditional formatting)
Sort / filter	"Sort by sales descending and filter to Technology"
Pivot & charts	"Create a pivot table of sales by category and year with YoY %"
Clean up	"Fix the inconsistent city names" / "Split full name into first and last name"
Analyse with Python	"Run a correlation between discount and profit and plot it" (uses Python in Excel)

STEP BY STEP **A safe Copilot workflow**

1. Open your workbook (saved in OneDrive) → **Home → Copilot**.
2. Select the Table (or let Copilot detect it).
3. Ask one thing at a time, with column names: *"Add a column margin_pct = profit / sales, formatted as a percentage."*
4. **Read the proposed formula** and its explanation before clicking **Insert column**.
5. Check three rows by hand, and one total against a plain `SUM`.
6. Undo (`Ctrl + Z`) if wrong, and rephrase with more detail.

The COPILOT() function

Microsoft 365 also provides a `COPILOT` **function** that sends text and cell ranges to the model and returns results into cells that **spill and recalculate**:

Excel formula

```
=COPILOT("Classify this customer review as Positive, Neutral or Negative", D2:D100)
=COPILOT("Summarise the main complaint in 6 words or fewer", D2)
=COPILOT("Extract the city name from this address", A2)
```

Use it for text tasks (classify, summarise, extract, translate, tag) over **small to moderate** ranges, not for arithmetic: use normal formulas for numbers. Results can change between recalculations and

each call consumes model capacity; availability, limits and licensing change, so check Microsoft's current documentation. Convert important results to values (copy → paste values) once checked.

Here is the same classify-sentiment idea applied to the product reviews dataset, with the (non-AI) ground-truth labels already in the file, so you can see how to *measure* an AI's accuracy:

Python

```
import pandas as pd
rev = pd.read_csv("product_reviews.csv")
print(rev.shape)
rev[["review_id", "product", "rating", "review_text", "sentiment"]].head(6)
```

OUTPUT

```
(1200, 5)
```

	review_id	product	rating	review_text	sentiment
0	1	Speaker	4	Very comfortable to use all day	positive
1	2	Smartwatch	5	Very comfortable to use all day	positive
2	3	Smartphone	4	Delivery was quick and packaging was neat. great value for money, totally worth it	positive
3	4	Earbuds	1	Battery drains too fast	negative
4	5	Laptop	2	Delivery was late and the box was damaged	negative
5	6	Laptop	2	Refund process is very slow. heats up a lot while charging	negative

6 rows × 5 columns

Python

```python
# How would you evaluate an AI classifier? Compare its labels with a trusted label set.
# Here a deliberately weak keyword rule plays the role of "the AI" so you can see the evaluation method.
# (These reviews are synthetic and templated, so real reviews will be much harder to classify.)
neg_words = ("stopped", "late", "poor", "crashing")          # a deliberately small word list: an imperfect classifier
pos_words = ("excellent", "great", "amazing", "premium")
def guess(t):
    t = t.lower()
    n = sum(w in t for w in neg_words); p = sum(w in t for w in pos_words)
    return "negative" if n > p else ("positive" if p > n else "neutral")
rev["pred"] = rev.review_text.apply(guess)
acc = (rev.pred == rev.sentiment).mean()
print(f"Agreement with trusted labels: {acc:.1%}\n")
print(pd.crosstab(rev.sentiment, rev.pred, rownames=["actual"], colnames=["predicted"]))
```

OUTPUT

```
Agreement with trusted labels: 56.0%

predicted  negative  neutral  positive
actual
negative        204      206         0
neutral           0      175         0
positive          0      322       293
```

The cross-tab (a **confusion matrix**, ML track Lesson 15) shows *where* a classifier goes wrong. Do the same with an AI column: take 100 reviews you labelled yourself, compare, and only then trust it on 10,000.

Gemini in Google Sheets and with Excel

- **Gemini in Google Sheets** (Workspace plans): a side panel that can *create tables, write formulas, summarise, generate charts and analyse trends*, plus an **`AI()` function** to run a prompt per row (classify, summarise, extract) similar to `COPILOT()`.
- **Gemini and other LLM add-ins for Excel**: some marketplaces offer add-ins or custom functions (for example `=GEMINI("prompt", A2)`) that call a model through **your own API key**. They are useful for batch text tasks, but you must protect the key, watch the **cost per call**, and avoid sending sensitive data.
- **Apps Script / Office Scripts**: AI can write the script; you review and run it.

COMMON MISTAKES **API-key hygiene for add-ins**

Never paste an API key into a shared workbook or a cell others can read; use the add-in's secure settings, set a **spending limit** on the key, and rotate it if exposed. A key in a shared file is a billing and security incident waiting to happen.

Claude, ChatGPT and other assistants with your files

Upload a CSV or XLSX to a general assistant that can **run code** (analysis/code-execution tools). It reads the file, writes Python, runs it, and shows charts and tables: closer to an analyst than a chatbot that only talks.

A good first message with a file:

Text

```
I've attached superstore_sales.csv (one row per order line, Jan 2023-Dec 2025, amounts in
rupees).
1) Profile the data: row count, columns and types, missing values, duplicates, date range.
2) Compute total sales, profit and margin by year and category.
3) Find the three biggest drivers of low margin; quantify each.
4) Show your Python code, and tell me which assumptions you made.
Do not guess: if something cannot be computed from the file, say so.
```

Typical good behaviour: it prints the profile, shows code, makes charts. What you still do: spot-check numbers, make sure it used the right columns, and ask for the code so you can re-run it yourself. Assistants that run code produce **reproducible** analysis; those that only "think" may fabricate numbers.

Assistant strengths	Check
Writing and explaining formulas (even LET, LAMBDA, XLOOKUP, array formulas)	Test on sample rows
Cleaning plans and Power Query M code	Preview the result; keep a backup
Building DAX measures	Validate against a pivot
Summarising and drafting commentary	Verify each number quoted
Designing dashboards and chart choices	Compare with your audience's needs

Python in Excel

=PY(turns a cell into a Python editor (Formulas → Insert Python). It uses **pandas, matplotlib, seaborn, statsmodels and scikit-learn**, runs in Microsoft's secure cloud (no local Python needed), and reads Excel ranges with `xl()`.

Python

```
df = xl("tblSales[#All]", headers=True)          # Excel Table → pandas DataFrame
df.groupby("region")["profit"].sum().sort_values(ascending=False)
```

Ask Copilot: *"Use Python to show the correlation between discount and profit and draw a scatter plot."* Behind the scenes it writes the =PY() code. Check what it computed against this handbook's result:

Python

```
sup = pd.read_csv("superstore_sales.csv")
print(f"Correlation discount vs profit margin (lines): {sup.discount.corr(sup.profit / sup.sales):.2f}")
print(sup.groupby(sup.discount.map(lambda d: f"{d:.0%}")).apply(lambda g: round(g.profit.sum() / g.sales.sum() * 100, 1)).rename("margin_%").to_string())
```

OUTPUT

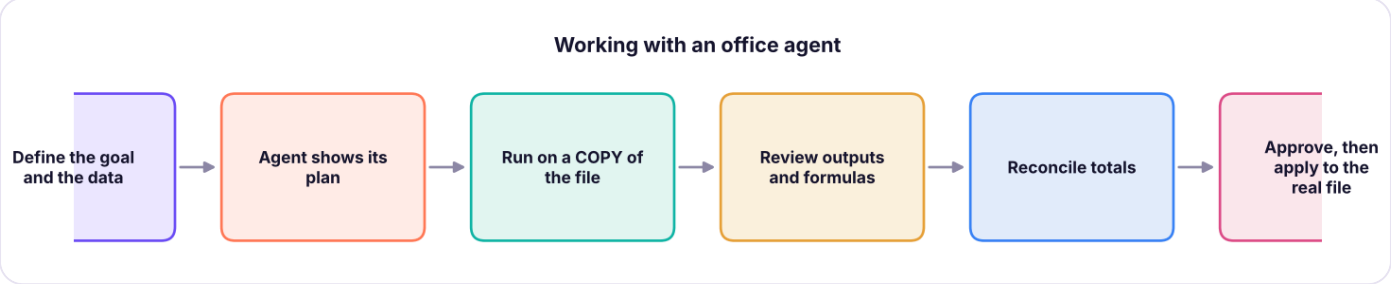
```
Correlation discount vs profit margin (lines): -0.91
discount
0%     17.5
10%     5.9
15%     0.9
20%    -4.6
30%   -19.0
40%   -26.3
```

A strong negative correlation, and margin turning negative from a 20% discount upwards: the same story as the SQL and Power BI lessons, arrived at a third way. Cross-checking tools against each other is how you build trust.

Office agents: the next step

Agents go beyond answering: given a goal ("Clean this export, build a pivot by region, add a chart, and draft a three-slide summary"), they **plan, use the spreadsheet or Office apps as tools, check results and iterate**, then present the outcome for your approval. Microsoft, Google, Anthropic, OpenAI and others have all released agent modes for office work; names and capabilities change quickly, so evaluate any specific product (including whichever "Frontier"-style agent your organisation provides) against the questions below.

Question to ask about any agent	Why it matters
What can it **read** and **write**? (files, mail, calendar, web)	Scope of risk
Does it **show its plan and steps** and ask before acting?	Oversight
Can I **undo** or review changes (version history)?	Recoverability
Where does data **go** (tenant, region, retention, training use)?	Compliance
How does it handle **untrusted content** (prompt injection in emails/web pages)?	Security
Does it produce **formulas and sources** or just pasted values?	Auditability
What does it **cost per task**?	Budget



A safe agent workflow: small steps, visible plan, human approval at risky points, independent checks.

What AI does well and badly in spreadsheets

Strong	Weak / risky
Explaining unfamiliar formulas and files	Large-scale precise calculation (use formulas/Python)
Drafting formulas, DAX, SQL, M code, VBA	Knowing your **business rules** unless told
Text tasks: classify, summarise, extract, translate	Consistent results every time (non-deterministic)
Suggesting charts and structure	Hidden assumptions (date ranges, definitions)
Debugging errors (`#N/A`, `#SPILL!`)	Anything involving **sensitive data** in the wrong tool
Documentation and commentary	Replacing review and sign-off

A checklist before you trust an AI-built spreadsheet

1. Totals reconcile with the source (`SUM` of raw vs summary).
2. Row counts match (no lost or duplicated rows).
3. Formulas are **visible** (not pasted values) and use absolute references correctly.
4. Spot-check five random rows by hand.

5. Edge cases: blanks, zero, negative, text in number columns, duplicates.
6. Units and currency are labelled; dates are real dates.
7. Assumptions are written on the sheet.
8. A second tool or person confirms the headline number.

Summary

KEY TAKEAWAYS

- Prepare data as **Tables** with clear headers, no merged cells, and a data dictionary; save in the cloud for Copilot.
- **Copilot** explains and writes formulas, creates pivots, charts, highlights and cleans data; `COPILOT()` and Sheets' `AI()` apply prompts per row for text tasks.
- General assistants that can **run code** make reproducible analyses from uploaded files; always request the code and assumptions.
- **Python in Excel** (`=PY()`) brings pandas into cells; AI can write it, you verify it.
- **Agents** plan and act across apps: work on a copy, require visible plans, approvals, undo, audit trail and clear data-handling terms.
- Verify with reconciliation, spot checks, edge cases and a second tool; protect keys and sensitive data.

Quick quiz

6 questions · tap an answer to check it

1. What should you do first to help an AI understand your Excel data?

A Merge headers

B Convert it to a clearly named Excel Table with descriptive headers and no merged cells

C Colour the cells

D Hide the columns

Answer: B. Structured, well-named data is easier for AI (and pivot tables) to use.

2. The COPILOT() function is best used for:

A Precise financial arithmetic

B Text tasks such as classifying, summarising or extracting over a range

C Replacing SUM

D Sorting

Answer: B. Use normal formulas for numbers; use AI functions for language tasks.

3. Why ask an assistant to show its Python code?

A To make the answer longer

B So the analysis is reproducible and can be checked

C Because code is always correct

D To avoid cleaning data

Answer: B. Visible code lets you verify and re-run the work.

4. Which is a sensible control for an office agent?

A Let it edit the live file without review

B Run it on a copy, review its plan and outputs, and keep version history

C Give it all passwords

D Disable logging

Answer: B. Human oversight and recoverability are essential.

5. An AI classifier labels reviews; how do you measure its quality?

A Read three results

B Compare its labels with a trusted labelled sample and look at accuracy and the confusion matrix

C Use temperature 1.0

D Ask it if it is right

Answer: B. Evaluate against ground truth, not self-reports.

6. Where should an API key for an Excel add-in be stored?

A In a visible cell

B In the add-in's secure settings with a spending limit

C In the sheet name

D In an email

Answer: B. Keys in shared files can be abused and billed to you.

EXERCISE

AI-assisted analysis, verified

Using `superstore_sales.csv`: (1) ask an AI assistant with code execution to profile the data and compute margin by category and discount band; (2) write the pivot table version yourself; (3) compare the two and list any differences; (4) create three `COPILOT()`-style prompts (or Gemini `AI()` prompts) for a column of customer comments, and say how you would evaluate them on 50 labelled examples.

Show solution (worked answer)

Expected margins by discount band: 0% → 17.5%, 1–20% → 1.2%, >20% → −20.6% (overall 9.1%). Differences usually come from averaging margins instead of total profit ÷ total sales, or from using order lines versus orders. Prompts: "Classify as Positive, Neutral or Negative (one word)"; "Extract the product mentioned (or 'none')"; "Summarise the complaint in under 8 words". Evaluation: label 50 comments yourself, run the prompt, compute accuracy and a confusion matrix, inspect errors, refine the prompt, and re-test on a fresh sample.

INTERVIEW CORNER

Questions you may be asked

Q: How would you use AI tools to speed up Excel analysis responsibly? Prepare structured data, ask specific questions, review generated formulas and pivots, reconcile totals, document assumptions, keep sensitive data out of unapproved tools, and keep a human sign-off.

Q: What is Python in Excel? A feature that runs Python (pandas, matplotlib, scikit-learn) in cells via `=PY()`, in a secure cloud, reading Excel ranges with `xl()`, useful for analysis and charts that are hard with formulas.

Q: What is an AI agent in an office context and what are its risks? A system that plans and executes multi-step tasks using apps and files; risks include wrong actions, data leakage, prompt injection and lack of audit; mitigate with scoped permissions, approvals, copies and logs.

Q: How do you check an AI-generated formula? Test on a few rows including edge cases, compare totals with a simple independent calculation, read the formula's explanation, and make sure references and units are right.

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